Assignment 2

June 9, 2026 4:48 PM

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Lab Assignment 2

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Lab Assignment #2 – Designing, Developing, and Architecting Interactive Android Applications with Material Design 3 and Advanced Features

Due Date: June 15, 2026 at 11:59 pm

Purpose: The purpose of this lab assignment is to:

- Apply Material Design 3 color schemes to create visually appealing and consistent app themes.
- Design responsive UIs that adapt to large screens and foldable devices using WindowSizeClass.
- Implement accessibility features to ensure apps are usable by all users, including those with disabilities.
- Implement the Model-View-ViewModel (MVVM) architecture to enhance code separation and maintainability.
- Create composable functions using LazyColumn to efficiently display lists.
- Utilize Jetpack libraries to streamline and enhance app development.
- Apply dependency injection to improve code modularity and testability.

References: Textbook, ppt slides, class examples, and Android tutorials (https://developer.android.com/develop/ui/views/theming/look-and-feel). This material provides the necessary information that you need to complete the exercises.

- Ensure to read the following general instructions carefully:
- This assignment must be **completed individually** by all the students.
- You will have to **demonstrate your solution in a scheduled lab session and upload the solution on Luminat**e through the assignment link under Assessments.

Exercise 1

You will develop a **Student Career Development Hub** application designed to help students organize and track their career growth activities. The app should focus on providing a smooth user experience with appealing UI/UX design and modern Android development practices. Students will be able to manage skills, certifications, projects, and career goals from a centralized platform.

The app must include the following functionalities:

- A dashboard/home screen displaying a list of ongoing career development activities, skills, certifications, or career goals with progress indicators.
- A screen to create a new career item (skill, certification, project, internship application, or career goal).
- A screen to view and edit existing career items.
- Responsive layouts for mobile and tablet use, with a focus on accessibility features.
- MVVM architecture to ensure clean code separation and easier maintenance. *atleast 3 *screens? files?
- Integration with Jetpack libraries, including ViewModel, StateFlow, Navigation, and dependency injection for scalable development.

Features and Implementations

1. Material Design 3

- Apply Material Design 3 principles and color schemes throughout the application.
- Use Material Design components such as AppBar, Buttons, Cards, TextFields, Dialogs, and Floating Action Buttons (FAB).

2. Responsive UI Design

- Implement responsive layouts using WindowSizeClass.
- Support various screen sizes, including phones, tablets, and foldable devices.

3. Accessibility

- Add accessibility features such as:
  - Content descriptions
  - Larger touch targets
  - Proper screen reader support
  - Accessible color contrast and navigation

4. App Architecture (MVVM)

- Implement the MVVM architecture.
- Create ViewModel classes for managing UI-related data.
- Use StateFlow to observe and manage UI state changes.
- Separate UI, business logic, and data layers.

5. Jetpack Compose and LazyColumn

- Use LazyColumn to efficiently display lists of career development items.
- Create reusable composable functions for list items and UI components.
- Apply state hoisting where appropriate.

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6. Jetpack Libraries and Dependency Injection

* coin or Hilt *

- Utilize Jetpack libraries.
- Implement dependency injection (Hilt or Koin).
- Use Room Database for local data persistence.
- Use Navigation Compose for screen navigation.

Activities and Navigation

1. Home Screen

Displays a list of career development items using a LazyColumn.

Each item displays:

- Title
- Category
- Progress Status
- Completion Indicator

Categories may include:

- Skill Development
- Certification
- Academic Project
- Internship Application
- Career Goal

Features:

- Floating Action Button (FAB) to navigate to the Create Career Item Screen.
- Search or filter options (optional).

2. Create Career Item Screen

Contains input fields for:

- title
- description
- category
    - Skill Development
    - Certification
    - Academic Project
    - Internship Application
    - Career Goal
- start_date
- target_completion_date
- status
    - Not Started
    - In Progress
    - Completed
- progress_percentage

Includes:

- Save button to save the item and return to the Home Screen.
- Validation for required fields.

3. View/Edit Career Item Screen

Displays the selected career item's details.

Allows the user to:

- View complete information
- Edit the item
- Update progress
- Change status

Includes:

- Save button to save updates and return to the Home Screen.

Evaluation table:

Item	Points	Details
Functionality:		
Correct implementation of MVVM architecture:		
		Ensure ViewModel classes manage UI-

Evaluation table:

Item	Points	Details
Functionality:		
Correct implementation of MVVM architecture:		
ViewModel classes and StateFlow	8 points	Ensure ViewModel classes manage UI-related data and StateFlow are used for observing data changes.
Correct implementation of Material Design 3:		
Use of Material Design components and color schemes	4 points	Apply Material Design components (AppBar, Buttons, Cards, TextFields) and color schemes.
Implementation of responsive UI and accessibility features:		
Responsive layouts using WindowSizeClass	3 points	Implement responsive UI layouts that adapt to different screen sizes, including foldable devices.
Accessibility features for usability	2 points	Add accessibility features such as content descriptions, larger touch targets, and proper navigation for screen readers.
Correct implementation of Jetpack Compose and LazyColumn:		
Use of LazyColumn for efficient list display	2 points	Implement LazyColumn for displaying lists efficiently.
Friendliness:		
Alignments of UI controls	2 points	UI controls should be properly aligned and organized, providing a visually appealing layout.
Friendly I/O	1 point	The app should provide a user-friendly interface with intuitive input/output operations.
Comments, Correct Naming of Variables, Methods, Classes, etc.	2 points	Code should be well-documented with appropriate comments. Variables, methods, and classes should follow proper naming conventions.
Total	24 points	

Android Module Naming rules:

You must name your Android Studio project according to the following rule:

yourfullname_COMP304Labnumber_Exercisenumber.

Example: **johnsmith_COMP304Lab2_Ex1**

Submission rules:

Submit your projects as zip files that are named according to the following rule:

yourfullname_COMP304Labnumber_Exercisenumber.zip

Example: **johnsmith_COMP304Lab2_Ex1.zip**

Use Android Studio Export to zip feature to zip your projects.